



Upper Garren RP sources

A report prepared on behalf of Wye Salmon Association (WSA)

This report summarises the water quality results obtained by Isabela Foy, for her A-level geography project, over the summer of 2023 in the upper Garren water body. The project was designed to further understand the origins of elevated reactive phosphorus concentrations identified by earlier WSA sampling in the sub catchment above Orcop.

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Introduction

The upper Garren waterbody fails to achieve good status for orthophosphate, whereas the Gamber and lower Garren waterbodies do¹ – if only marginally so in the latter case. Moreover, the long-term trend on EA water quality data (Green, 2022) suggests that the Upper Garren is getting worse. This begs the question why? What is going on?

In recent years, EA monitoring of the upper Garren has been sporadic and, when it has occurred, has been largely limited to Llangunnock Bridge at the lower end of this waterbody. However, WSA Citizen Science (CS) water quality data, presented in Green, 2022, suggested that a large part of the problem originates in the headwaters above Orcop. Annual average RP at Orcop was found to be 0.10 mgP/l, compared to 0.07 mgP/l at Llangunnock Bridge. Moreover, a tributary stream above Orcop (we refer to this below as Brooklands Stream since it joins the Garren Brook adjacent to Brooklands Farm) was found to have annual average RP at 0.19 mgP/l.

Further work was clearly required, and an opportunity presented itself over the summer of 2023. A sixth-former, Isabela Foy, approached WSA looking for an A-level geography project involving water quality issues. Thus, a plan was hatched for Isabela to temporarily join WSA CS activity to look specifically at the Garren above Orcop in more detail.

Water quality data obtained at this level of granularity will inevitably implicate individual farms. However, this has been done in the hope that landowners will be amenable to fixing any problems identified and the work is not intended to embarrass those concerned. It is simply following the data.

The catchment

The Garren Brook catchment above Orcop, shown in figure 1 below, occupies approximately 975 ha (red outline). The brook rises on the eastern slopes of Garway Hill and flows eastward towards Orcop. From Low Moors Farm at the foot of Garway Hill to Orcop the brook flows a total distance of 3.25 km. Elevation falls some 40m over the 1.9 km between Low Moors Farm and Brooklands Farm. From Brooklands to Orcop, things are more sedate, with a fall of only 5m.

Agriculture is predominantly pasture (cattle & sheep) with smaller pockets of arable activity, as detailed in Table 1.

Land use	Fraction
Improved grassland	62%
Arable	27%
Broadleaved woodland	7%
Suburban	3%

Table 1: land use in the Garren catchment above Orcop

There are in the region of a dozen farms within the catchment, the largest of these (based of farm building area) being shown in Figure 1. Of these, three are notable for other activities:

- Burhope Farm has an IPU housing 32,000 free-range egg layers, increased from 16,000 in 2017. Poultry litter has been spread on the land (largely pasture) since 2008. The range area is immediately adjacent to a Garren tributary. The farm also hosts camping facilities for about 30 caravans, complete with shower and toilet block, and a small fishing lake.
- Brooklands Farm has an IPU housing 16,000 free-range egg layers, spreading litter on the land since 2002. An AD plant (250kW) was added in 2014, feed stock being poultry litter plus 5,000

¹ 2019 WFD assessment

tonnes per year of arable crops. Digestate is spread on the land growing the arable feedstock, so effectively this land is still receiving surplus phosphorus equal to the poultry load.

- Great Ynys Farm has three poultry sheds housing 90k broilers, in operation since 2000. An AD plant (190kW) was added in 2010, feedstock being poultry litter, some imported cattle manure, plus 3,300 tonnes per year of maize, grown on the farm.

Feedstock for AD at Brooklands and Great Ynys accounts for around one third of the arable land within this sub catchment.

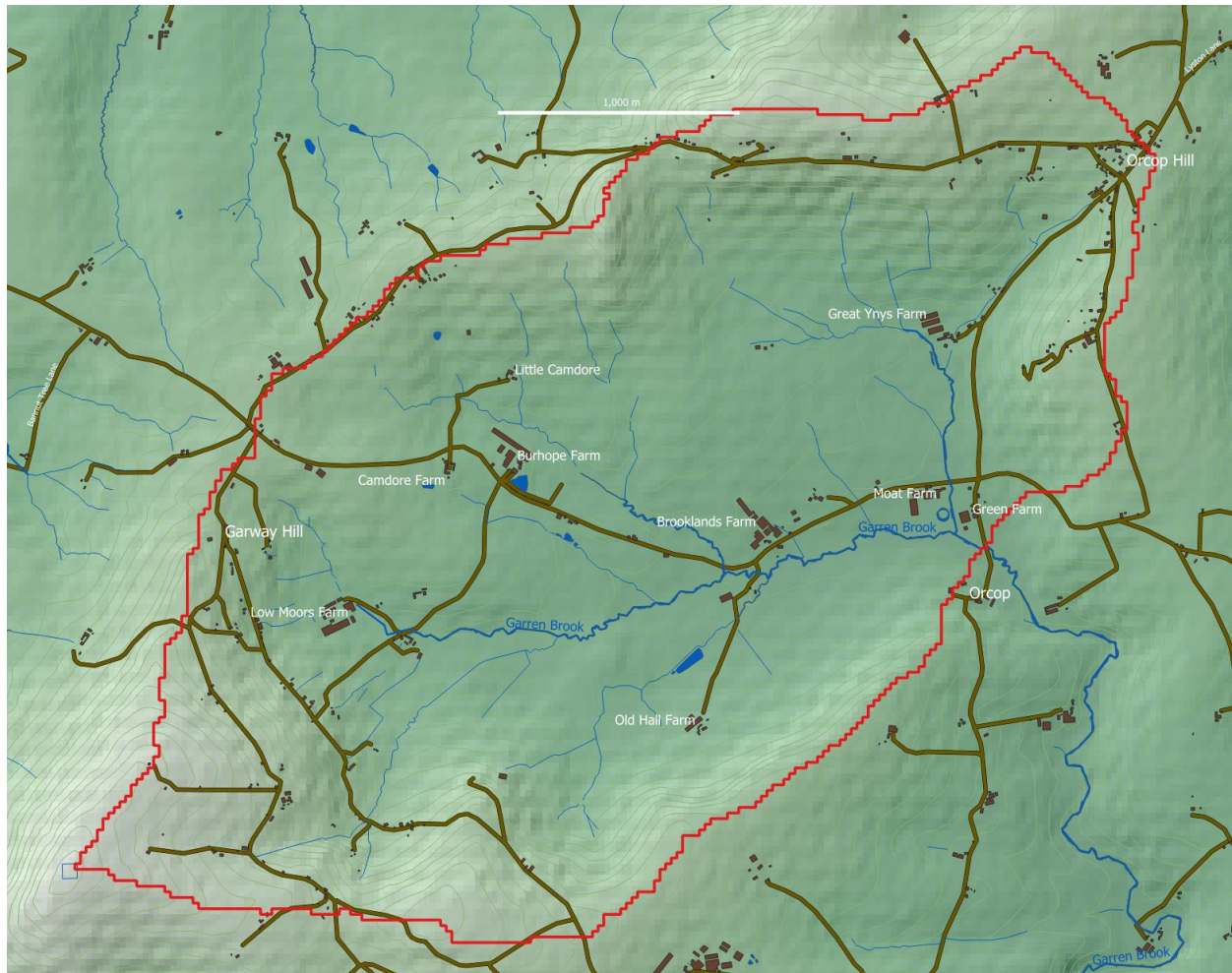


Figure 1: topography of the Garren sub catchment above Orcop. Red outline is the sub catchment boundary. Some of the larger farms are also indicated.

Sampling

The project duration of ten weeks would not allow the generation of ‘annual average’ statistics, nevertheless ten weekly samples is sufficient to generate statistically robust comparative data, comparing site A to site B. As the summer of 2023 unfolded, with rather mixed weather, we have ended up with a dataset spanning a wide range of flow conditions, and which is not entirely unrepresentative of annual average data.

Within the constraints of safety of access, seven sites were selected. These are shown in Figure 2 below. Five of these are at road bridges and two (F1 & F2) are on a public footpath. The rationale for the locations is as follows:

- Point A, at Orcop bridge is the sub catchment outlet and replicates the location used in earlier WSA CS work (Green, 2022). It is also immediately downstream of the Great Ynys Stream confluence.
- Point B, a new location, samples a stream draining Great Ynys Farm, in addition to some other farms further up-slope. We shall refer to this stream as the Great Ynys Stream.
- Point C, a new location, monitors the Garren immediately downstream of the Brooklands Stream confluence.
- Point D replicates the second location used in earlier WSA CS work. It samples the Brooklands Stream immediately above the Garren confluence.
- Point E, a new location, lies immediately below Low Moors Farm and is intended to indicate the 'background level' of RP, resulting from 'typical' agricultural activities.
- Point F1, a new location, lies immediately above a confluence on the Brooklands Stream, on what we shall refer to as the eastern fork. This stream, and the Brooklands Stream below this confluence together form the boundary between Brooklands Farm and Burhope Farm. The east fork thus drains part of both of these farms, plus some other land further up-slope.
- Point F2, a new location, lies on the western fork of Brooklands stream, draining Burhope Farm, Little Camdore and further up-slope land. It passes immediately behind the Burhope IPU - the long thin building visible in figure 2. The poultry range area appears to lie between this building and the west fork.

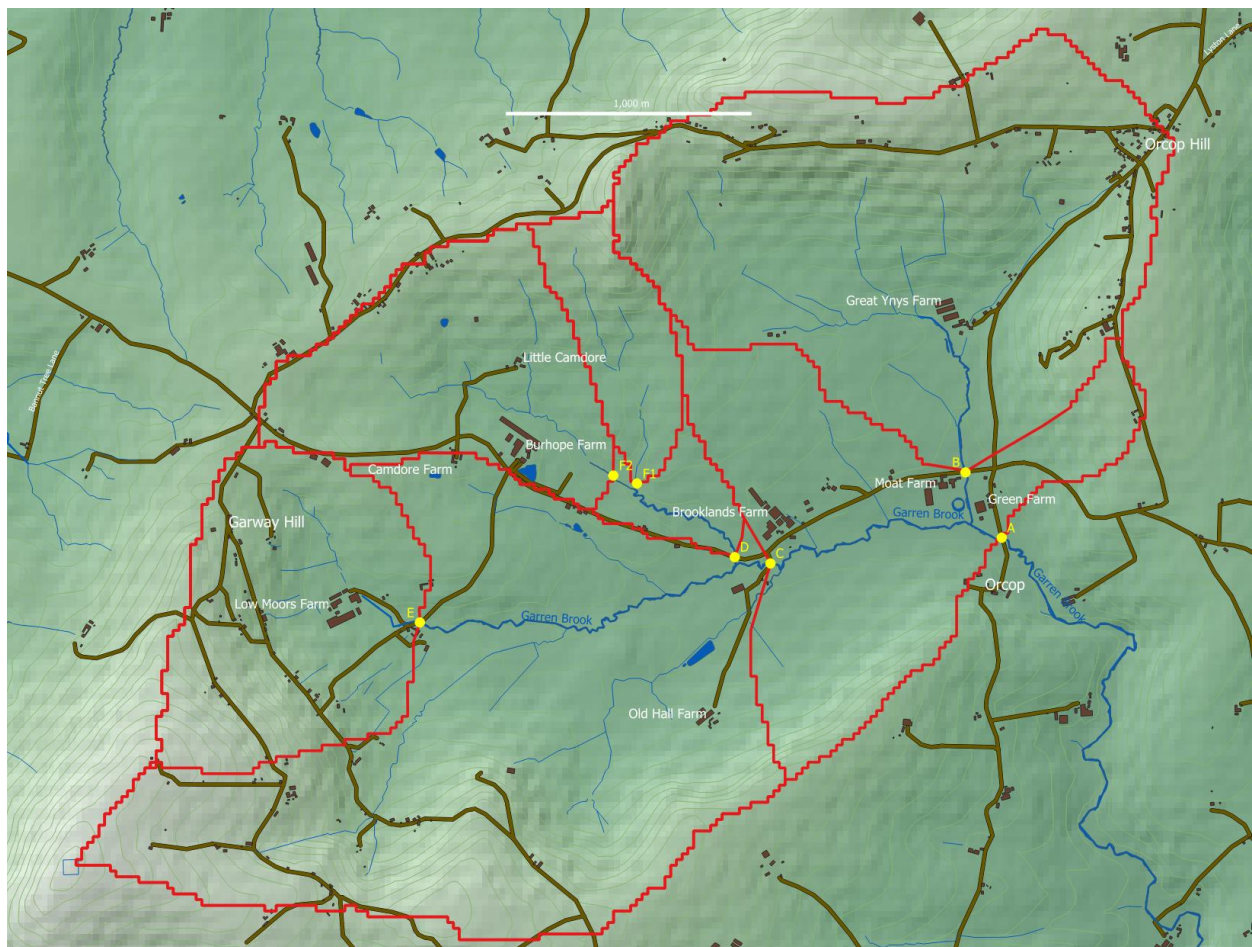


Figure 2: location of sampling points for this work. Points A & D replicate the locations used in earlier work, the remainder are all new. The red boundaries show the catchments above each sampling point.

Sampling at each of these seven locations occurred weekly, for ten weeks. RP was measured using a Hanna Low-range Phosphate Checker. Other parameters were measured but are not reported here.

Prior work

Key points from Green, 2022 are as follows:

- Orcop (Point A) has an average RP concentration of 0.101 mgP/l (for reference, the WFD limit for good status at Orcop, taking account of elevation and alkalinity, is 0.067 mgP/l). Moreover, this is substantially higher than further downstream at Llangunnock Bridge (0.071 mgP/l).
- Brooklands Stream (point D) had an annual average RP concentration of 0.192 mgP/l. Furthermore, LAM analysis for this location showed a strong fixed flow component, approximating to the sewage from 42 people (if that is indeed what it is).

Results & discussion

Average results from this work are shown in table 2, and below in figure 3 as a catchment schematic.

Location	Mean RP (mgP/l)
A: Orcop Bridge	0.11
B: Garren below Great Ynys confluence	0.19
C: Garren below Brooklands confluence	0.14
D: Brooklands Stream above Garren conf.	>0.28
E: Garren below Low Moors	0.12
F1: Brooklands east fork	0.12
F2: Brooklands west fork	>0.40

Table 2: average RP concentration by location. Note that the F2 result is shown as greater than because 3 out of 10 results were off scale on the Hanna instrument (>0.82 mgP/l). Point D likewise had 1 off scale reading.

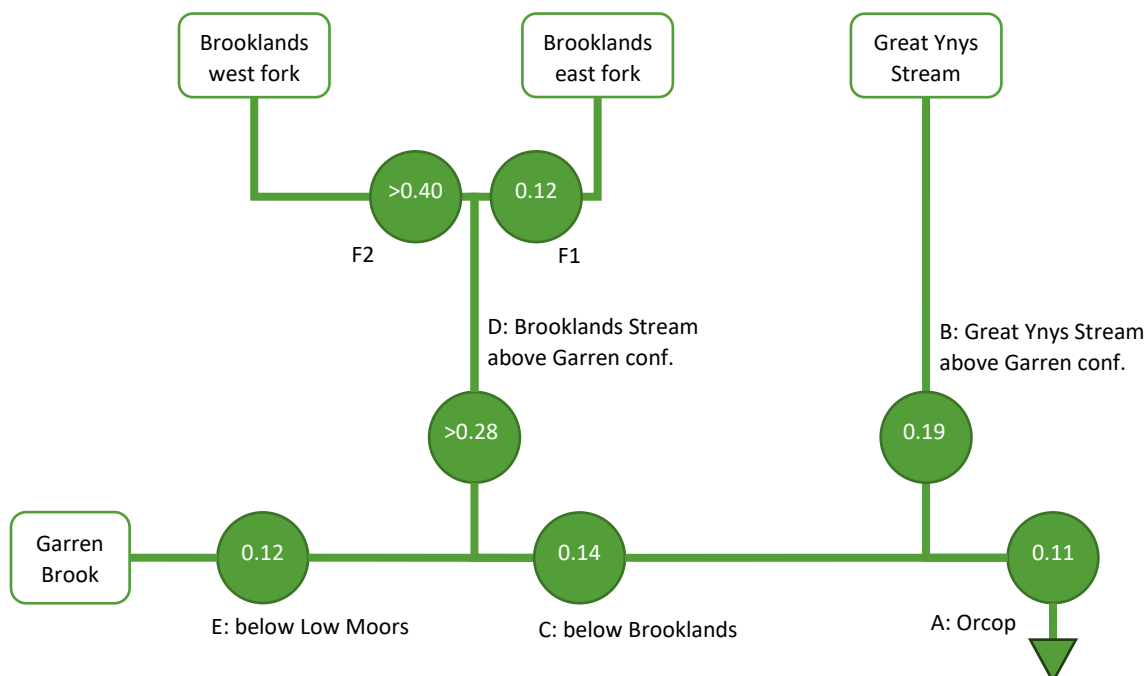


Figure 3: catchment schematic showing mean RP as mgP/l.

Point E shows us that, within half a kilometre of its source, the Garren is already overloaded with phosphorus – with almost twice the WFD compliance limit. Land use above point E is 12% woodland, 13% arable and 67% improved grassland. Thus, it is hard to avoid the conclusion that the high RP is a

symptom of over-fertilised pasture. Indeed, there is also a possibility that some of this land is also receiving poultry litter from Burhope, or digestate from Brooklands; this is not known.

At point A, we have a value which is very similar to the annual average reported in Green, 2022. And, very similar to Point E. Likewise, point F1 on the east fork of Brooklands Stream shows a similar value.

Point B draining Great Ynys, on the other hand, is at three times the WFD compliance limit. Something anomalous is clearly going on in this area. We can correlate the data for this location with the recorded water height parameter to get some sense of its flow dependence, as shown in figure 4.

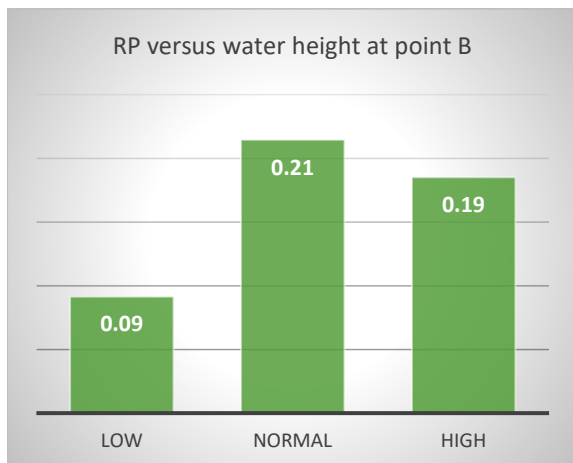


Figure 4: average RP for different (subjective) water height assessments on the Great Ynys Stream.

This is rather suggestive of diffuse pollution, with smaller RP values at low water height.

Point F2 has truly eye-watering levels of RP. In fact, given that the Hanna instrument overloaded on 3 of the 10 readings, we can assume that the true value is somewhat higher than the 0.40 mgP/l reported above. The results for F1 and F2 show quite clearly that the gross problem identified in Green, 2022 is caused by water flowing from Burhope Farm in the west fork of Brooklands stream. As above, we can examine the water height correlation:

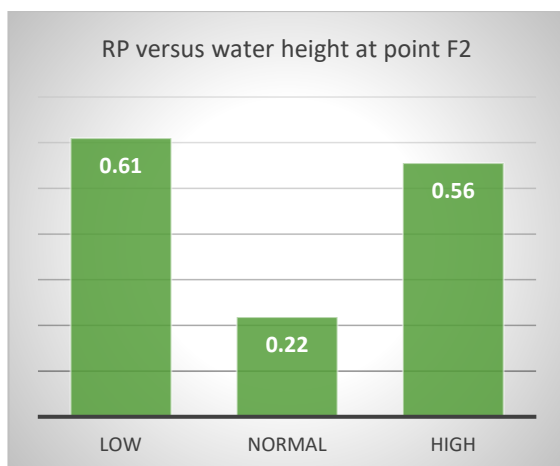


Figure 5: average RP for different (subjective) water height assessments on the west fork of the Brooklands Stream

This shows very high values occurring in low flow conditions, indicative of fixed flow pollution, in addition to the diffuse component. This result is consistent with the LAM analysis for point D presented in Green, 2022. Moreover, despite the smaller sample sizes in this study, a comparison of the *Low* and *Normal* samples using a Student t-test shows 90% confidence in the result.

Finally, it is useful to consider RP mass flowrates within this sub catchment. The areas of the individual catchment areas shown in figure 2 can be used to estimate the stream discharge (volumetric flowrate) at the sampling locations. This allows us to calculate the corresponding RP mass flowrate.

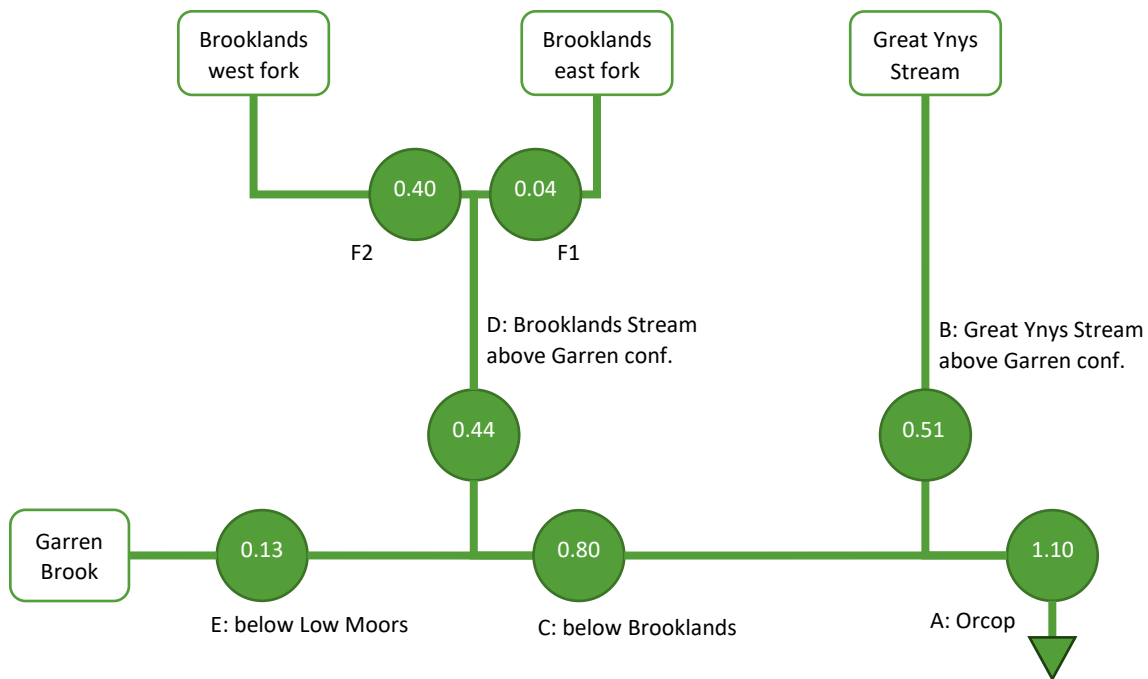


Figure 6: catchment schematic showing RP flow as kgP/day.

Figure 6 shows very clearly how two streams, passing through Burhope Farm and Great Ynys Farm respectively, are emitting huge amounts of RP relative to the rest of the catchment. The lower RP value found on the Great Ynys Stream belies the fact that this stream has approximately twice the discharge of the Brooklands Stream and, thus, contributes a greater mass flow into the Garren. Clearly, these results do not prove beyond doubt that these farms are indeed the sources (there being other farms within the respective catchments). But it does form rather compelling circumstantial evidence which deserves to be investigated.

Figure 6 also illustrates one difficulty of monitoring phosphorus using RP. The signal decays in longer stretches of river as orthophosphate is modified and metabolised within the water column. Thus figure 6 does not everywhere obey the conservation of mass law. On Brooklands Stream it does, because all three points are physically close together. So, $F1 + F2 = D$. But distances E-C and C-A are sufficient for losses to occur. Including for the remaining catchment between E and A (using the per hectare export rate calculated for the catchment above point E) suggests that point C should be 0.94 kgP/day and point A should be 1.62 kgP/day. Using the latter number then, about a quarter of the phosphorus arriving at Orcop flows from Burhope Farm and almost a third of it flows from Great Ynys Farm. There are in the region of a dozen farms in this catchment and two, which happen to have IPUs, appear to account for 56% of the phosphorus export.

Conclusions

- The elevated RP levels found in the Garren Brook at Orcop are explained as follows:
 - There is a generally high 'background' level of RP (~0.12 mgP/l), presumably driven by common land management practices on pasture – this being the dominant sector in the area. This level of RP is approximately 80% above WFD good status.
 - Two specific streams, coincidentally both draining lands associated with long-term poultry litter spreading, show levels of RP well in excess of this:
 - Brooklands Stream west fork (SO 45895 26674), passing through Burhope Farm, had a mean RP concentration in excess of 0.40 mgP/l – six times the limit for WFD good status. Moreover, this location suffers from a fixed flow load, in addition to the expected diffuse pollution. Whilst this contributes only around 10% of the annualised RP mass flow, it nevertheless becomes the dominant contributor in low flow conditions.
 - Great Ynys Stream (SO 47332 26687), passing through Great Ynys Farm, had a mean RP concentration of 0.19 mgP/l – almost three times the limit for WFD good status.
 - These two streams (and potentially two farms) are contributing 56% of the RP export within the sub catchment above Orcop. This excess RP mass flow is significant to the whole of the upper Garren water body. Eliminating it could potentially bring the waterbody into WFD good status.
- Further sampling of these streams, say upstream of suspected sources, requires access to private land and is beyond the scope of citizen science work. In addition to additional water quality checks, there is clearly a pressing need to understand soil phosphorus levels in these areas.
- We therefore request that the Environment Agency investigate the above streams as a matter of urgency and enforce appropriate remediation.

List of abbreviations

CS	Citizen Science
LAM	Load Apportionment Modelling
RP	Reactive Phosphorus
WFD	Water Framework Directive
WSA	Wye Salmon Association

References

Green, G. (2022). *Catchment Overview - Garren Brook*.